

Clinical Trial Survival Analysis of Adjuvant Chemotherapy

A study of time to recurrence in colon cancer using R

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1. Introduction

In clinical studies, it is important to know not only whether a patient experiences a recurrence, but also **when** it occurs.

For example, two patients may both experience recurrence, but one may experience it after 400 days while another experiences it after 900 days. Survival analysis is useful because it considers this time information.

Another important issue is **censoring**. Some patients complete their follow-up without experiencing recurrence. Their exact recurrence time is therefore unknown.

This project uses survival-analysis methods in R to study the time to recurrence in colon cancer clinical-trial data.

2. Objective

The main objective is:

To study time to cancer recurrence across treatment groups and examine whether treatment and selected patient characteristics are associated with recurrence hazard.

The analysis focuses on:

- Treatment group
- Time to recurrence
- Censoring
- Age
- Sex
- Number of positive lymph nodes

3. Dataset

The analysis uses the `colon` dataset from the R `survival` package.

For this project, records with:

`etype = 1`

are selected to focus on recurrence.

Variable	Meaning
rx	Treatment group
time	Follow-up time
status	Recurrence / censoring indicator
age	Patient age
sex	Patient sex
nodes	Number of positive lymph nodes

The treatment groups are Observation (`Obs`), Levamisole (`Lev`), and Levamisole + 5-FU (`Lev+5FU`).

4. Analysis Workflow

Clinical Trial Data → Data Preparation → Kaplan–Meier → Log-Rank Test → Cox Model → PH Assumption

5. Data Preparation

The recurrence records are first selected and the variables required for the analysis are retained.

```
library(survival)

data(colon)

recurrence <- subset(colon, etype == 1)

recurrence <- recurrence[, c(
  "rx", "time", "status",
  "age", "sex", "nodes"
)]
```

The survival outcome is represented using:

`Surv(time,status)`

where `time` gives the follow-up time and `status` indicates whether recurrence was observed.

6. Kaplan–Meier Analysis

The Kaplan–Meier method estimates the probability that a patient remains free from recurrence as follow-up time increases.

The curve decreases when recurrence events occur. Censored observations do not directly decrease the curve.

Main question:

How does recurrence-free survival change over time for the different treatment groups?

R Code

```
surv_obj <- Surv(
  recurrence$time,
  recurrence$status
)

km_model <- survfit(
  surv_obj ~ rx,
  data = recurrence
)

plot(km_model)
```

Kaplan–Meier Result

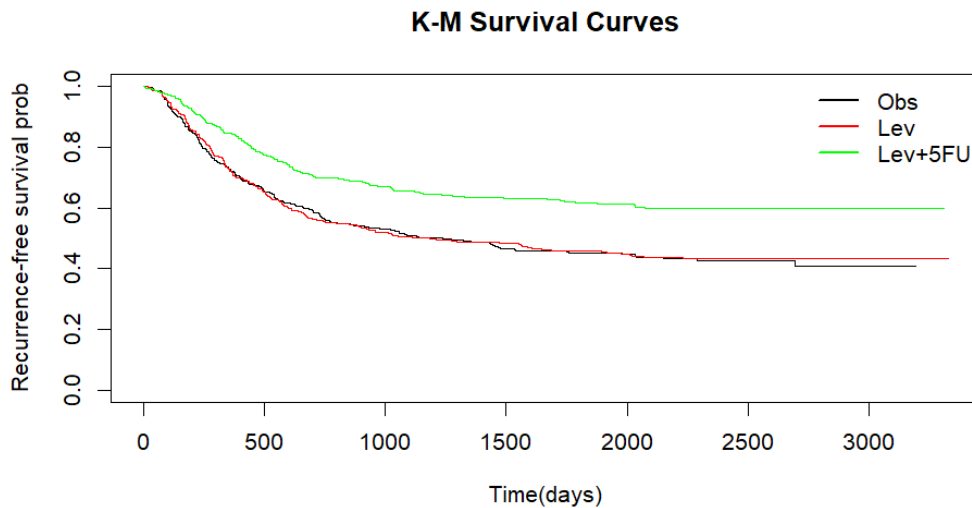


Figure 1: Kaplan–Meier recurrence-free survival curves by treatment group.

Interpretation:

Replace this text after generating your actual graph.

For example:

The Kaplan–Meier curves show how recurrence-free survival changes over time across the treatment groups. The separation between curves provides a visual indication of differences in recurrence patterns.

Do not claim that one treatment is effective based only on visual separation. The formal comparison is performed using the log-rank test.

7. Log-Rank Test

The log-rank test provides a formal comparison of the survival curves.

```
survdif(  
  Surv(time, status) ~ rx,  
  data = recurrence  
)
```

The hypotheses are:

H_0 : The treatment groups have the same survival distribution

H_1 : At least one treatment group differs

Result

Add your actual:

- Chi-square statistic
- Degrees of freedom
- p-value

Log-rank result:

$\chi^2 = 23.1$
 $df = 2$
 $p = 0.00001$

Interpretation:

8. Cox Proportional Hazards Model

The Cox model is used to examine the relationship between recurrence hazard and treatment or patient characteristics. The model used is:

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)$$

In simple terms, the model compares the recurrence hazard while considering several variables together.

R Code

```
cox_model <- coxph(  
  Surv(time, status) ~  
    rx + age + sex + nodes,  
  data = recurrence  
)  
  
summary(cox_model)  
  
exp(coef(cox_model))  
  
exp(confint(cox_model))
```

Hazard Ratio

The hazard ratio is calculated as:

$$HR = e^{\beta}$$

A simple interpretation is:

- $HR = 1$: no estimated difference in hazard
- $HR < 1$: lower estimated hazard relative to the reference
- $HR > 1$: higher estimated hazard relative to the reference

The reference treatment group should be clearly stated in the final results.

Cox Model Results

Replace the values below with the actual output from R.

Variable	HR	Lower 95%	Upper 95%	p-value
Levamisole	–	0.7541	1.1550	0.765
Lev + 5-FU	–	0.4592	0.7367	0.765
Age	–	0.9887	1.0042	0.988
Sex	–	0.7146	1.0337	0.086
Nodes	–	1.0676	1.1053	0.443

Table 1: Cox proportional hazards model results.

Interpretation:

The hazard ratios show the estimated association of each variable with recurrence hazard while holding the other variables in the model constant.

9. Proportional Hazards Assumption

The Cox model assumes that the relative hazard between groups remains approximately stable over time. This is called the **proportional hazards assumption**.

It is checked using:

```
ph_test <- cox.zph(cox_model)
ph_test
plot(ph_test)
```

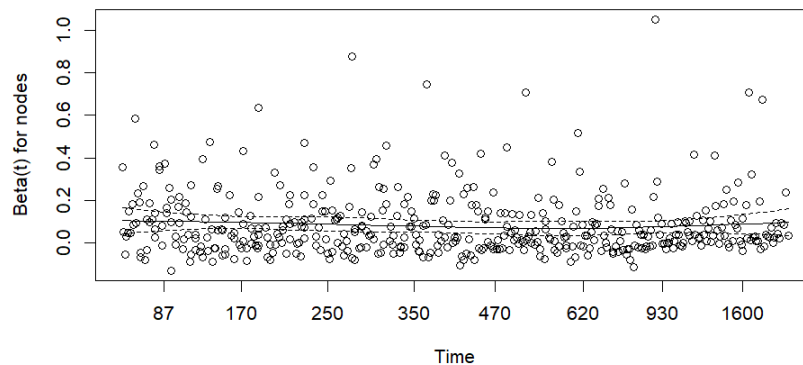


Figure 2: Proportional hazards assumption check.

Result:

Add the actual output from `cox.zph()` here.

The proportional hazards assumption was assessed using Schoenfeld residuals. The conclusion should be based on the actual test results and the corresponding diagnostic plots.

10. Main Findings

After inserting the actual R results, summarize the project in 3–4 points:

- **Kaplan–Meier:** Describe the main pattern observed in recurrence-free survival.
- **Log-rank test:** State whether the treatment-group survival curves showed statistically significant evidence of a difference.
- **Cox model:** Mention the important hazard-ratio findings for treatment and patient characteristics.
- **Model assumption:** State the conclusion from the proportional hazards check.

11. Conclusion

This project demonstrates the use of survival-analysis methods in R for clinical trial data.

Kaplan–Meier analysis describes recurrence-free survival over time, the log-rank test compares treatment-group survival curves, and Cox regression examines the association between recurrence hazard and treatment or patient characteristics.

The analysis also demonstrates the importance of checking model assumptions before interpreting the final Cox model.

The results should be interpreted as statistical associations within the analyzed data and study design rather than automatically as proof of a causal treatment effect.

12. Tools

Software: R / RStudio

Main package: survival

Methods: Kaplan–Meier, Log-Rank Test, Cox Proportional Hazards Regression